TORRENS UNIVERSITY AUSTRALIA

ASSESSMENT 2 BRIEF	
Subject Code and Title	MLN 601 Machine Learning
Assessment	Classification
Individual/Group	Individual
Length	1,500 words (+/–10%) and 7–10 min Presentation
Learning Outcomes	The Subject Learning Outcomes demonstrated by successful completion of the task below include: b) Apply learning algorithms to perform machine learning tasks. c) Implement practical machine learning: data pre-processing, analysis, model selection, and interpret the results. d) Communicate clearly and effectively using the technical language of machine learning to a range of stakeholders
Submission	Due by 11.55 pm AEST/AEDT on the Sunday at the end of Module 8
Weighting	40%
Total Marks	100 marks

Task Summary

Assessment 1 considered a wine data set as a regression task. This brief revisits the data set as a classification task. In this Assessment, you will use a decision tree Machine Learning (ML) algorithm to analyse data and draw conclusions. To help you create and document this ML model and the results, you will follow the end-to-end CRoss-Industry Standard Process for Data Mining (CRISP-DM) (Chapman et al., 2000) methodology. Further, to guide you through the analysis, the development of your report and model, and the writing of your report and 7–10 minute presentation, a template for your Jupyter Notebook has been provided with comments. Your presentation should touch on the key steps of the template, including the lessons you learned and your experiences.

Please refer to the Task Instructions (below) for further details on how to complete this task.

Context

In addition to giving you an opportunity to complete a ML exercise, this Assessment also gives you an opportunity to practice hyperparameter tuning using the really useful scikit-learn library. In your future workplaces, you will often be expected to perform similar exercises using suitable data sets with different machine learners and tune the hyperparameters. Model building requires you to revise parameters and tune them for the next model run.

For this Assessment, the practice data set is available from the UCI ML repository, which contains nearly 500 real-world data sets (https://archive.ics.uci.edu/ml/index.php). Your focus will be on the wine quality data set (https://archive.ics.uci.edu/ml/datasets/wine+quality). This data set provides wine quality data across 11 traits, including acidity, residual sugar and alcohol concentration. Importantly, this Assessment requires you to develop a model to predict wine quality on a score between 1 to 10.

You will revisit this data set to complete a classification task. To achieve this, you will setup a categorical variable with two categories. Thus, you will be required to allocate levels for your wine quality (the dependent variable) to assign either a 'low' quality (1) (below the value of 6) or a 'high' quality (0) (below the selected value of 6). You will use this binary classification to help generate a prediction model for high or low quality wine using decision tree algorithms.

Follow the steps of the CRISP-DM model using the template CRISP_DM _Template_(assessment_2_ classification).ipynb to document and develop your ML model. At the modelling stage, you should practice tuning the hyperparameters for the decision tree to ascertain the effects on the model and determine the optimal performance using the AUC-ROC curve.

Task Instructions

You will use your Jupyter Notebook on the Microsoft Azure ML platform or Google Colab and Python 3.6 as the language for all three assessments.

Ultimately, the Notebook will contain both your ML code, data and report documentation.

Your Assessment will be evaluated based on the major stages of the CRISP-DM process as set out in the Notebook template with prompts. The process comprises:

1. Business Understanding;
2. Data Understanding;
3. Data Preparation;
4. Modelling;
5. Evaluation; and
6. Deployment.

The six multi-step stages of the CRISP-DM must be undertaken to complete this Assessment. Note: For ease of working and to complete this Assessment, you should document what you are doing in your Notebook as you progress through the activities (e.g., the steps undertaken and the rationale for the selection of the code). The template will prompt you on how to work through the end-to-end ML process.

Stage 1: Business Understanding

1. This section serves as an introduction. You should write a clear and concise narrative expressing what you are trying to achieve. Think in terms of ML; for example, the prediction algorithm, the data set selected, what you are seeking from the data set and how you intend to understand the value of your prediction capability.
2. Assess the current situation. See 1.1 of the CRISP-DM template (1.1).

Stage 2: Data Understanding

1. Acquire the relevant wine quality data set from the UCI repository for your prediction model (https://archive.ics.uci.edu/ml/datasets/wine+quality). Explicitly specify the data source by providing a specific link and the name of the data set (e.g., red wine, white wine or both) and the method of acquisition (e.g., direct from the URL or a download of the .csv file). The steps taken need to be clearly stated. (2.1).
2. Read this data set into your Notebook. (2.1).
3. Describe the data set inclusive of variables, units and levels. (2.2).
4. Verify the data quality by analysing the data set for structure and missing data. (2.3).
5. Conduct an initial data exploration using data visualisation, reporting and querying of the data. (2.4).
6. Use the pairplot function in seaborn to determine the relationship, if any, between the variables. Include the output or the visualisation of the pairplot function in your Notebook and comment on it. (2.4.2).

Stage 3: Data Preparation

1. Select the data that you will use for the analysis. (3.1).
2. Clean the data you have selected to improve the quality of the data. (3.2).

Stage 4: Modelling

1. For this Assessment, you are only required to consider one classification modelling technique (e.g., a decision tree).
2. Import the decision tree model in your code. (4.1).
3. Record any modelling assumptions. (4.2).
4. Run your model over the data set. (4.3).
5. Record the parameter settings, your rationale for your choice of values and the actual model generated. (4.3).
6. Revise any parameter settings for subsequent model runs. Document all the revisions until the best model is reached. (4.4).
7. Assess the model or models according to the performance measurement set to meet your evaluation criteria. The AUC-ROC curve is useful for the performance measurement of classification.
8. Revise any parameter settings for subsequent model runs. Document all the revisions until the best model is reached. (4.4).

Stage 5: Evaluation

1. Assess the ML results. Ensure you include a statement as to whether the model meets the initial objective.

Stage 6: Deployment

1. For this Assessment, you are not required to deploy your model. For this stage, simply include any lessons that you learned and that you wish to share in relation to the things that went right and wrong, the areas in which you did well and in which you could improve. You can also detail any of your other experiences in completing this Assessment.

Stage 7: Presentation

1. Once complete, you should setup the Jupyter Notebook for screen recording. You are required to make a screen recording of your Jupyter Notebook and a webcam video of yourself narrating for 7–10 minutes. You should specify your name and any other student details at the beginning. Work your way through the Notebook as you discuss the key aspects of the CRISP-DM steps, the lessons you learned and any other experiences.

2. A wide variety of tools are available to record videos of a webcam and screen simultaneously (i.e., picture in picture). In this case, your video will show you discussing your Notebook on your screen. Use the large screen for your Notebook. Available tools include the inbuilt recorder for Windows 10, Quicktime on Apple, fluvid.com, panopto.com or Zoom. Owing to the size of the video file, you will be submitting the URL for the file. Practice the presentation beforehand to ensure clarity and conciseness.

References

Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C. & Wirth, R. (2000). *CRISP-DM 1.0—Step-by-step data mining guide*. Copenhagen, Denmark: SPSS Inc. https://www.the-modeling-agency.com/crisp-dm.pdf

Cortez, P., Cerdeira, A., Almeida, F., Matos, T. & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems, Elsevier*, *47*(4), 547–553. https://archive.ics.uci.edu/ml/data sets/wine+quality

UCI (2019). *Machine learning repository*. Irvine, CA: University of California, School of Information and Computer Science. http://archive.ics.uci.edu/ml

Submission Instructions

You are expected to submit three files: 1) The Notebook file with an extension .ipynb; 2) the direct PDF download via LaTeX from the Notebook; 3) the URL. Name the files as follows:

MLN601LastNameFirstNameBrief2.ipynb;

MLN601LastNameFirstNameBrief2.pdf; and

[URL to your filename] MLN601LastNameFirstNameBrief2.mp4.

IMPORTANT NOTE: Submit a text file (.txt) that includes the code(s) of your programs in your submissions.

Submit the files and the URL via the Assessment link in the main navigation menu in **MLN601—Machine Learning**.

The Learning Facilitator will provide feedback via the Grade Centre in the LMS portal. Feedback can be viewed in My Grades.

Academic Integrity Declaration

I declare that except where I have referenced, the work I am submitting for this assessment task is my own work. I have read and am aware of Torrens University Australia Academic Integrity Policy and Procedure viewable online at http://www.torrens.edu.au/policies-and-forms

I am aware that I need to keep a copy of all submitted material and their drafts, and I will do so accordingly.

Assessment Rubric

Assessment Attributes	Fail (Yet to Achieve Minimum Standard) 0–49%	Pass (Functional) 50–64%	Credit (Proficient) 65–74%	Distinction (Advanced) 75–84%	High Distinction (Exceptional) 85–100%
Business Understanding 5%	Not really attempted, incomplete and difficult to comprehend.	Demonstrates an attempt to frame the problem (e.g., to predict red wine quality).	Contains a clear explanation of the motivation to undertake ML and the problem statement.	This entire introduction to the report is clear.	Complete, readable and easy to understand.
Data Understanding 15%	Entirely incorrect and not completely attempted. Totally lacking any interpretation.	Partially correct but missing plots and poorly labelled. Provides limited comments on the source of the and variables.	Plots and analysis partially complete, including the correlation analysis. Minor omissions in interpretation.	The analysis describing the data and plots is complete and includes a visualisation of the relationships between the attributes.	The data understanding has been explicitly tied to the data analysis and the context. All the plots convey information and include adequate references to the data set.
Data Preparation 10%	Comments missing and appears to be incomplete.	Partially complete.	Covers data quality and discusses the data cleaning steps that can be undertaken.	The discussion of the data preparation is mostly clear.	The discussion is clear and appears to be correct with some depth. Some additional commentary suggestions. For example, the missing data can be estimated.

Modelling and Evaluation **20%**	Overly simplistic and incomplete attempt at modelling.	An appropriate analysis of the ML code but fails to include model assessment comments.	The modelling generation and assessment are mainly correct but include minor errors or omissions. Attempt to review performance using AUC-ROC with some errors.	The modelling is complete and mainly correct in relation to the performance review but includes minor errors or lacks some depth.	The modelling is complete and includes an assessment and informative comments that aid interpretation.
Deployment **10%**	No real attempt to draw a conclusion detailing the lessons learned or experiences (no or very little reflection).	The conclusion is relatively well structured but fails to detail the lessons learned or discuss the experience.	The conclusion details the lessons learned but not much more.	The conclusion details the lessons learned and discusses the experience.	The conclusion clearly details the lessons learned and discusses the experience. The conclusion has some depth (e.g., suggests what could be done better next time).
Deliverables I: • **Jupyter code** • **Pdf** **25%**	Insufficient comments made in the markdown reporting. Video, Jupyter Notebook file or pdf not provided. Did not attempt to use the template provided. The video did not use the Notebook extension to present or improvise.	Notebook file requires major changes to run. Markdown is hard to follow and confusing. The presentation is similarly hard to follow as it is dependent on the markdown.	Notebook runs with minor changes. Markdown is reasonable and can be followed. Pdf output is similar to the Notebook. The video and narrative moves from beginning to	All files run and the Notebook reproduces the submitted file. The presentation follows the Notebook slides and the narrative is clear.	The markdown report documents the entire end-to-end process and makes full use of the CRISP-DM stages as highlighted in the template. The Notebook file runs with no errors and matches the pdf. The presentation touches all key stages of the CRISP-

			share lessons learned and experiences at the end.		DM and concludes with a discussion of the student's own experiences.
Deliverables II: • **Video presentation** **15%**	Speaks in a low volume and/or monotonous tone, which causes the audience to disengage. Does not introduce her/himself or clearly define the walk through of the Notebook that will take place or the purpose; provides weak or no support of the modelling undertaken as a classification; provides insufficient support of the lessons learned and/or experiences. Fails to increase audience understanding or their knowledge of the topic.	Speaks in an uneven volume with little or no inflection. Attempts to define the purpose and the subject; provides irrelevant examples, facts and and/or statistics, which do not adequately support the subject; includes very thin data or evidence. Little or no attempt made to walk through the stages of the CRISP-DM as conducted for the Assessment. Increases audience understanding and knowledge of at least the most important stages of the template, including model selection.	Speaks in an uneven volume with little inflection. Attempts to define the purpose and the subject; provides weak examples, facts and and/or statistics, which do not adequately support the Notebook; includes very thin data or evidence of following the template. Increases audience understanding and knowledge of some aspects of the development of the Notebook especially the key stages as defined in the template provided.	Speaks with satisfactory variation in volume and inflection. Has a somewhat clear purpose and the subject is somewhat clear; includes some examples, facts and and/or discusses the tests and statistics that support the modelling; includes some data or evidence that supports the lessons learned and experiences. Increases audience understanding and awareness of the key stages indicated in the template.	Speaks with fluctuation in volume and inflection to maintain audience interest and emphasise key points. Provides a clear purpose and details of the Notebook; includes pertinent examples, facts and and/or statistics; supports conclusions/ideas with evidence by walking through the Notebook. Significantly increases audience understanding and knowledge of the topic; convinces the audience to recognise the validity and importance of the Notebook, computations, explanations and lessons learned.

The following Subject Learning Outcomes are addressed in this assessment	
SLO b)	Apply learning algorithms to perform machine learning tasks.
SLO c)	Implement practical machine learning: data pre-processing, analysis, model selection and interpret the results.
SLO d)	Communicate clearly and effectively using the technical language of machine learning to a range of stakeholders